

OLDACHE OUSSAMA NAZIM

Marine Geoscientist | Remote Sensing, Bathymetry & Coastal Engineering

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PROFESSIONAL PROFILE

Recent MSc graduate in Marine Geosciences and Coastal Engineering (USTHB, June 2026), building on an applied geology background that included sedimentary and coastal field placements. Comfortable across the full workflow, from field data acquisition through Sentinel-2 satellite image processing to GIS analysis and numerical coastal modelling. Research work focused on satellite-derived bathymetry and marine pollution monitoring, with rigorous statistical model validation. Seeking a first professional role within a technical team on coastal, offshore or environmental projects.

TECHNICAL SKILLS

- Marine geology and sedimentology
- Coastal geomorphology and nearshore dynamics
- Coastal erosion and sediment transport
- Bathymetry and seabed analysis
- Remote sensing and satellite imagery (Sentinel-2)
- Geographic Information Systems (GIS)
- Geospatial data processing and spatial analysis
- Numerical coastal modelling (waves, sediments)

EDUCATION

MSc (Master II) — Marine Geosciences and Coastal Engineering

USTHB, Houari Boumediene University of Science and Technology, Algiers • June 2026

BSc (Licence) — Applied Geology

USTHB, Houari Boumediene University of Science and Technology, Algiers • 2023

ACADEMIC PROJECTS

MSc Thesis — Coupled Physicochemical and Remote Sensing Analysis of Marine Pollution, Bay of Algiers

USTHB, Master II • 2025 – 2026

- Combined in-situ physicochemical measurements with Sentinel-2 imagery to characterise coastal marine pollution
- Computed and interpreted spectral indices for monitoring marine environmental parameters
- Carried out satellite image processing, spatial analysis and statistical validation of results
- Produced mapping of marine environmental parameters across the Bay of Algiers

Satellite-Derived Bathymetry (SDB) — Les Sablettes Beach

USTHB, applied research project

- Built a complete processing pipeline, from Sentinel-2 pre-processing in Google Earth Engine through to machine learning modelling
- Benchmarked Stumpf, Lyzenga and Random Forest approaches for depth retrieval from spectral variables
- Cross-checked predictions against in-situ bathymetric survey data; random and spatial validation assessed via R^2 , RMSE and MAE
- Delivered bathymetric maps, a technical report and an interactive topo-bathymetric viewer

Modelling Coursework and Field Placements

- Wave propagation modelling with SWAN; sediment transport and shoreline evolution analysis
- Coastal risk assessment and protection solutions: breakwaters, groynes, beach nourishment, nature-based solutions
- Field placement in marine geosciences and coastal engineering — Tigzirt
- Field placement in applied geology, sedimentary formations — Bou Saâda

SOFTWARE & TOOLS

GIS & Remote Sensing: ArcGIS Pro, QGIS, Google Earth Engine, ESA SNAP, DSAS, Sentinel-2 data

Numerical Modelling: SWAN

Programming & Data: Python, MATLAB, JavaScript (basic)

Machine Learning & Statistics: Random Forest, statistical analysis, model validation, spatial analysis

LANGUAGES

Arabic — native

French — B2

English — B2